

The Sunflower Lemma of Erdős and Rado

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Abstract

We formally define sunflowers and provide a formalization of the sunflower lemma of Erdős and Rado: whenever a set of size- k -sets has a larger cardinality than $(r - 1)^k \cdot k!$, then it contains a sunflower of cardinality r .

1 Sunflowers

Sunflowers are sets of sets, such that whenever an element is contained in at least two of the sets, then it is contained in all of the sets.

theory *Sunflower*

imports *Main*

HOL-Library.FuncSet

begin

definition *sunflower* :: 'a set set $\Rightarrow$ bool **where**

sunflower $S = (\forall x. (\exists A B. A \in S \wedge B \in S \wedge A \neq B \wedge$
 $x \in A \wedge x \in B)$
 $\longrightarrow (\forall A. A \in S \longrightarrow x \in A))$

lemma *sunflower-subset*: $F \subseteq G \Longrightarrow \text{sunflower } G \Longrightarrow \text{sunflower } F$

unfolding *sunflower-def* **by** *blast*

lemma *pairwise-disjnt-imp-sunflower*:

pairwise disjnt $F \Longrightarrow \text{sunflower } F$

unfolding *sunflower-def*

by (*metis disjnt-insert1 mk-disjoint-insert pairwiseD*)

lemma *card2-sunflower*: **assumes** *finite S* **and** $\text{card } S \leq 2$

shows *sunflower S*

proof –

from *assms* **have** $\text{card } S = 0 \vee \text{card } S = \text{Suc } 0 \vee \text{card } S = 2$ **by** *linarith*

with $\langle \text{finite } S \rangle$ **obtain** $A B$ **where** $S = \{\}$ $\vee S = \{A\}$ $\vee S = \{A, B\}$

using *card-2-iff[of S]* *card-1-singleton-iff[of S]* **by** *auto*

thus *?thesis* **unfolding** *sunflower-def* **by** *auto*

qed

```

lemma empty-sunflower: sunflower {}
  by (rule card2-sunflower, auto)

lemma singleton-sunflower: sunflower {A}
  by (rule card2-sunflower, auto)

lemma doubleton-sunflower: sunflower {A,B}
  by (rule card2-sunflower, auto, cases A = B, auto)

lemma sunflower-imp-union-intersect-unique:
  assumes sunflower S
    and  $x \in (\bigcup S) - (\bigcap S)$ 
  shows  $\exists! A. A \in S \wedge x \in A$ 
proof -
  from assms obtain A where A:  $A \in S$   $x \in A$  by auto
  show ?thesis
  proof
    show  $A \in S \wedge x \in A$  using A by auto
    fix B
    assume B:  $B \in S \wedge x \in B$ 
    show  $B = A$ 
    proof (rule ccontr)
      assume  $B \neq A$ 
      with A B have  $\exists A B. A \in S \wedge B \in S \wedge A \neq B \wedge x \in A \wedge x \in B$  by auto
      from  $\langle \text{sunflower } S \rangle$  [unfolded sunflower-def, rule-format, OF this]
      have  $x \in \bigcap S$  by auto
      with assms show False by auto
    qed
  qed
qed

lemma union-intersect-unique-imp-sunflower:
  assumes  $\bigwedge x. x \in (\bigcup S) - (\bigcap S) \implies \exists_{\leq 1} A. A \in S \wedge x \in A$ 
  shows sunflower S
  unfolding sunflower-def
proof (intro allI impI, elim exE conjE, goal-cases)
  case (1 x C A B)
  hence x:  $x \in \bigcup S$  by auto
  show ?case
  proof (cases x ∈ ∩ S)
    case False
      with assms[of x] x have  $\exists_{\leq 1} A. A \in S \wedge x \in A$  by blast
      with 1 have False unfolding Uniq-def by blast
      thus ?thesis ..
    next
      case True
      with 1 show ?thesis by blast
  qed

```

qed

lemma *sunflower-iff-union-intersect-unique*:

sunflower $S \longleftrightarrow (\forall x \in \bigcup S - \bigcap S. \exists! A. A \in S \wedge x \in A)$
 (is ?l = ?r)

proof

assume ?l

from *sunflower-imp-union-intersect-unique*[OF this]

show ?r by auto

next

assume ?r

hence *: $\forall x \in \bigcup S - \bigcap S. \exists_{\leq 1} A. A \in S \wedge x \in A$

unfolding *ex1-iff-ex-Uniq* by auto

show ?l

by (rule *union-intersect-unique-imp-sunflower*, insert *, auto)

qed

lemma *sunflower-iff-intersect-Uniq*:

sunflower $S \longleftrightarrow (\forall x. x \in \bigcap S \vee (\exists_{\leq 1} A. A \in S \wedge x \in A))$
 (is ?l = ?r)

proof

assume ?l

from *sunflower-imp-union-intersect-unique*[OF this]

show ?r unfolding *ex1-iff-ex-Uniq*

by (metis (no-types, lifting) *DiffI UnionI Uniq-I*)

next

assume ?r

show ?l

by (rule *union-intersect-unique-imp-sunflower*, insert <?r>, auto)

qed

If there exists sunflowers whenever all elements are sets of the same cardinality r , then there also exists sunflowers whenever all elements are sets with cardinality at most r .

lemma *sunflower-card-subset-lift*: **fixes** $F :: 'a \text{ set set}$

assumes *sunflower*: $\bigwedge G :: ('a + \text{nat}) \text{ set set}.$

$(\forall A \in G. \text{finite } A \wedge \text{card } A = k) \implies \text{card } G > c$

$\implies \exists S. S \subseteq G \wedge \text{sunflower } S \wedge \text{card } S = r$

and kF : $\forall A \in F. \text{finite } A \wedge \text{card } A \leq k$

and $\text{card}F$: $\text{card } F > c$

shows $\exists S. S \subseteq F \wedge \text{sunflower } S \wedge \text{card } S = r$

proof –

let ?n = $\text{Suc } c$

from $\text{card}F$ have $\text{card } F \geq ?n$ by auto

then obtain FF where sub : $FF \subseteq F$ and $\text{card}F$: $\text{card } FF = ?n$

by (rule *obtain-subset-with-card-n*)

let ?N = $\{0 \dots ?n\}$

from $\text{card}F$ have *finite* FF

by (*simp add: card-ge-0-finite*)

```

from ex-bij-betw-nat-finite[OF this, unfolded cardF]
obtain f where f: bij-betw f ?N FF by auto
hence injf: inj-on f ?N by (rule bij-betw-imp-inj-on)
have Ff: FF = f ‘ ?N
  by (metis bij-betw-imp-surj-on f)
define g where g = ( $\lambda i. (Inl \text{ ‘ } f i) \cup (Inr \text{ ‘ } \{0 ..< (k - card (f i))\})$ )
have injg: inj-on g ?N unfolding g-def using f
proof (intro inj-onI, goal-cases)
  case (1 x y)
    hence f x = f y by auto
    with injf 1 show x = y
      by (meson inj-onD)
qed
hence cardgN: card (g ‘ ?N) > c
  by (simp add: card-image)
{
  fix i
  assume i  $\in$  ?N
  hence f i  $\in$  FF unfolding Ff by auto
  with sub have f i  $\in$  F by auto
  hence card (f i) ≤ k finite (f i) using kF by auto
  hence card (g i) = k ∧ finite (g i) unfolding g-def
    by (subst card-Un-disjoint, auto, subst (1 2) card-image, auto intro: inj-onI)
}
hence  $\forall A \in g \text{ ‘ } ?N. finite A \wedge card A = k$  by auto
from sunflower[OF this cardgN]
obtain S where SgN: S ⊆ g ‘ ?N and sf: sunflower S and card: card S = r
by auto
from SgN obtain N where NN: N ⊆ ?N and SgN: S = g ‘ N
  by (meson subset-image-iff)
from injg NN have inj-g: inj-on g N
  by (rule inj-on-subset)
from injf NN have inj-f: inj-on f N
  by (rule inj-on-subset)
from card-image[OF inj-g] SgN card
have cardN: card N = r by auto
let ?S = f ‘ N
show ?thesis
proof (intro exI[of - ?S] conjI)
  from NN show ?S ⊆ F using Ff sub by auto
  from card-image[OF inj-f] cardN show card ?S = r by auto
  show sunflower ?S unfolding sunflower-def
  proof (intro allI impI, elim exE conjE, goal-cases)
    case (1 x C A B)
      from  $\langle A \in f \text{ ‘ } N \rangle$  obtain i where i: i  $\in$  N and A: A = f i by auto
      from  $\langle B \in f \text{ ‘ } N \rangle$  obtain j where j: j  $\in$  N and B: B = f j by auto
      from  $\langle C \in f \text{ ‘ } N \rangle$  obtain k where k: k  $\in$  N and C: C = f k by auto
      hence gk: g k  $\in$  g ‘ N by auto
      from  $\langle A \neq B \rangle$  A B have ij: i  $\neq$  j by auto

```

```

    from inj-g ij i j have gij:  $g\ i \neq g\ j$  by (metis inj-on-contrad)
    from  $\langle x \in A \rangle$  have memi:  $Inl\ x \in g\ i$  unfolding A g-def by auto
    from  $\langle x \in B \rangle$  have memj:  $Inl\ x \in g\ j$  unfolding B g-def by auto
    have  $\exists A\ B. A \in g \wedge N \wedge B \in g \wedge N \wedge A \neq B \wedge Inl\ x \in A \wedge Inl\ x \in B$ 
      using memi memj gij i j by auto
    from sf[unfolded sunflower-def SgN, rule-format, OF this gk] have  $Inl\ x \in g$ 
k .
    thus  $x \in C$  unfolding C g-def by auto
  qed
qed
qed

```

We provide another sunflower lifting lemma that ensures non-empty cores. Here, all elements must be taken from a finite set, and the bound is multiplied the cardinality.

lemma *sunflower-card-core-lift*:

```

assumes finE: finite (E :: 'a set)
and sunflower:  $\bigwedge G :: 'a\ set\ set. (\forall A \in G. finite\ A \wedge card\ A \leq k) \implies card\ G > c$ 
 $\implies \exists S. S \subseteq G \wedge sunflower\ S \wedge card\ S = r$ 
and F:  $\forall A \in F. A \subseteq E \wedge s \leq card\ A \wedge card\ A \leq k$ 
and cardF:  $card\ F > (card\ E\ choose\ s) * c$ 
and s:  $s \neq 0$ 
and r:  $r \neq 0$ 
shows  $\exists S. S \subseteq F \wedge sunflower\ S \wedge card\ S = r \wedge card\ (\bigcap S) \geq s$ 
proof -
  let ?g =  $\lambda (A :: 'a\ set). card\ A = s \wedge A \subseteq E$ 
  let ?E =  $\{X. X \subseteq E \wedge card\ X = s\}$ 
  from cardF have finF: finite F
    by (metis card.infinite le-0-eq less-le)
  from cardF have FnE:  $F \neq \{\}$  by force
  {
    from FnE obtain B where B:  $B \in F$  by auto
    with F[rule-format, OF B] obtain A where A:  $A \subseteq E \wedge card\ A = s$ 
      by (meson obtain-subset-with-card-n order-trans)
    hence ?E  $\neq \{\}$  using B by auto
  } note EnE = this
  define f where f =  $(\lambda A. SOME\ x. ?g\ A\ x)$ 
  from finE have finiteE: finite ?E by simp

  have  $f \in F \rightarrow ?E$ 
  proof
    fix B
    assume B:  $B \in F$ 
    with F[rule-format, OF B] have  $\exists x. ?g\ B\ x$  by (meson obtain-subset-with-card-n)
    from someI-ex[OF this] B F show  $f\ B \in ?E$  unfolding f-def by auto
  qed
  from pigeonhole-card[OF this finF finiteE EnE]
  obtain a where a:  $a \in ?E$ 

```

```

    and le: card F ≤ card (f - ' {a} ∩ F) * card ?E by auto
  have precond: ∀ A ∈ f - ' {a} ∩ F. finite A ∧ card A ≤ k
    using F finite-subset[OF - finE] by auto
  have c * (card E choose s) = (card E choose s) * c by simp
  also have ... < card F by fact
  also have ... ≤ (card (f - ' {a} ∩ F)) * card ?E by fact
  also have card ?E = card E choose s by (rule n-subsets[OF finE])
  finally have c < card (f - ' {a} ∩ F) by auto
  from sunflower[OF precond this]
  obtain S where *: S ⊆ f - ' {a} ∩ F sunflower S card S = r
    by auto
  from finite-subset[OF - finF, of S]
  have finS: finite S using * by auto
  from * r have SnE: S ≠ {} by auto
  have finIS: finite (⋂ S)
  proof (rule finite-Inter)
    from SnE obtain A where A: A ∈ S by auto
    with F s have finite A
      using * precond by blast
    thus ∃ A ∈ S. finite A using A by auto
  qed
  show ?thesis
  proof (intro exI[of - S] conjI *)
    show S ⊆ F using * by auto
    {
      fix A
      assume A ∈ S
      with *(1) have A ∈ f - ' {a} and A: A ∈ F using * by auto
      from this have **: f A = a A ∈ F by auto
      from F[rule-format, OF A] have ∃ x. card x = s ∧ x ⊆ A
        by (meson obtain-subset-with-card-n order-trans)
      from someI-ex[of ?g A, OF this] **
      have a ⊆ A unfolding f-def by auto
    }
    hence a ⊆ ⋂ S by auto
    from card-mono[OF finIS this]
    have card a ≤ card (⋂ S) .
    with a show s ≤ card (⋂ S) by auto
  qed
qed

lemma sunflower-nonempty-core-lift:
  assumes finE: finite (E :: 'a set)
  and sunflower: ⋀ G :: 'a set set.
    (∀ A ∈ G. finite A ∧ card A ≤ k) ⇒ card G > c
    ⇒ ∃ S. S ⊆ G ∧ sunflower S ∧ card S = r
  and F: ∀ A ∈ F. A ⊆ E ∧ card A ≤ k
  and empty: {} ∉ F
  and cardF: card F > card E * c

```

```

shows  $\exists S. S \subseteq F \wedge \text{sunflower } S \wedge \text{card } S = r \wedge (\bigcap S) \neq \{\}$ 
proof (cases  $r = 0$ )
  case False
    from F empty have  $F': \forall A \in F. A \subseteq E \wedge 1 \leq \text{card } A \wedge \text{card } A \leq k$  using finE
    by (metis One-nat-def Suc-leI card-gt-0-iff finite-subset)
    from cardF have  $\text{card } F': (\text{card } E \text{ choose } 1) * c < \text{card } F$  by auto
    from sunflower-card-core-lift[OF finE sunflower, of k c F 1, OF - - F' cardF' - False]
    obtain S where  $S \subseteq F$  and main:  $\text{sunflower } S \wedge \text{card } S = r \wedge 1 \leq \text{card } (\bigcap S)$  by
    auto
    thus ?thesis by (intro exI[of - S], auto)
  next
    case True
    thus ?thesis by (intro exI[of - {}], auto simp: empty-sunflower)
qed

end

```

2 The Sunflower Lemma

We formalize the proof of the sunflower lemma of Erdős and Rado [2], as it is presented in the textbook [3, Chapter 6]. We further integrate Exercise 6.2 from the textbook, which provides a lower bound on the existence of sunflowers.

```

theory Erdos-Rado-Sunflower
imports
  Sunflower
begin

```

When removing an element from all subsets, then one can afterwards add these elements to a sunflower and get a new sunflower.

```

lemma sunflower-remove-element-lift:
  assumes  $S: S \subseteq \{ A - \{a\} \mid A \in F \wedge a \in A \}$ 
  and sf: sunflower S
  shows  $\exists Sa. \text{sunflower } Sa \wedge Sa \subseteq F \wedge \text{card } Sa = \text{card } S \wedge Sa = \text{insert } a \text{ ` } S$ 
proof (intro exI[of - insert a ` S] conjI refl)
  let  $?Sa = \text{insert } a \text{ ` } S$ 
  {
    fix B
    assume  $B \in ?Sa$ 
    then obtain C where  $C: C \in S$  and  $B: B = \text{insert } a \text{ ` } C$ 
    by auto
    from  $C \in S$  obtain T where  $T \in F \wedge a \in T \wedge C = T - \{a\}$ 
    by auto
    with B have  $B = T$  by auto
    with  $\langle T \in F \rangle$  have  $B \in F$  by auto
  }

```

```

thus  $SaF$ :  $?Sa \subseteq F$  by auto
have inj: inj-on (insert  $a$ )  $S$  using  $S$ 
  by (intro inj-on-inverseI[of -  $\lambda B. B - \{a\}$ ], auto)
thus  $\text{card } ?Sa = \text{card } S$  by (rule card-image)
show sunflower  $?Sa$  unfolding sunflower-def
proof (intro allI, intro impI)
  fix  $x$ 
  assume  $\exists C D. C \in ?Sa \wedge D \in ?Sa \wedge C \neq D \wedge x \in C \wedge x \in D$ 
  then obtain  $C D$  where *:  $C \in ?Sa \wedge D \in ?Sa \wedge C \neq D \wedge x \in C \wedge x \in D$ 
  by auto
  from  $*(1-2)$  obtain  $C' D'$  where
    **:  $C' \in S \wedge D' \in S \wedge C = \text{insert } a \ C' \wedge D = \text{insert } a \ D'$ 
  by auto
  with  $\langle C \neq D \rangle$  inj have  $CD'$ :  $C' \neq D'$  by auto
  show  $\forall E. E \in ?Sa \longrightarrow x \in E$ 
  proof (cases  $x = a$ )
    case False
    with * ** have  $x \in C' \wedge x \in D'$  by auto
    with **  $CD'$  have  $\exists C D. C \in S \wedge D \in S \wedge C \neq D \wedge x \in C \wedge x \in D$  by
auto
    from sf[unfolded sunflower-def, rule-format, OF this]
    show ?thesis by auto
  qed auto
qed
qed

```

The sunflower-lemma of Erdős and Rado: if a set has a certain size and all elements have the same cardinality, then a sunflower exists.

```

lemma Erdos-Rado-sunflower-same-card:
  assumes  $\forall A \in F. \text{finite } A \wedge \text{card } A = k$ 
  and  $\text{card } F > (r - 1) \wedge k * \text{fact } k$ 
  shows  $\exists S. S \subseteq F \wedge \text{sunflower } S \wedge \text{card } S = r \wedge \{\} \notin S$ 
  using assms
proof (induct k arbitrary: F)
  case 0
  hence  $F = \{\{\}\} \vee F = \{\}$   $\text{card } F \geq 2$  by auto
  hence False by auto
  thus ?case by simp
next
  case (Suc k F)
  define pd-sub :: 'a set  $\Rightarrow \text{nat} \Rightarrow \text{bool}$  where
    pd-sub =  $(\lambda G t. G \subseteq F \wedge \text{card } G = t \wedge \text{pairwise disjoint } G \wedge \{\} \notin G)$ 
  show ?case
  proof (cases  $\exists t G. \text{pd-sub } G \ t \wedge t \geq r$ )
    case True
    then obtain  $t G$  where pd-sub: pd-sub  $G \ t$  and  $t: t \geq r$  by auto
    from pd-sub[unfolded pd-sub-def] pairwise-disjnt-imp-sunflower
    have *:  $G \subseteq F \wedge \text{card } G = t \wedge \text{sunflower } G \wedge \{\} \notin G$  by auto
    from  $t \wedge \text{card } G = t$  obtain  $H$  where  $H \subseteq G \wedge \text{card } H = r$ 

```



```

    by (metis obtain-subset-with-card-n)
  with sunflower-subset[OF  $\langle H \subseteq G \rangle$ ] * show ?thesis by blast
next
case False
define P where P = ( $\lambda t. \exists G. \text{pd-sub } G t$ )
have ex:  $\exists t. P t$  unfolding P-def
  by (intro exI[of - 0] exI[of - {}], auto simp: pd-sub-def)
have large':  $\bigwedge t. P t \implies t < r$  using False unfolding P-def by auto
hence large:  $\bigwedge t. P t \implies t \leq r$  by fastforce
define t where t = (GREATEST t. P t)
from GreatestI-ex-nat[OF ex large, folded t-def] have Pt: P t .
from Greatest-le-nat[of P, OF - large]
have greatest:  $\bigwedge s. P s \implies s \leq t$  unfolding t-def by auto
from large'[OF Pt] have tr:  $t \leq r - 1$  by simp
from Pt[unfolded P-def pd-sub-def] obtain G where
  cardG: card G = t and
  disj: pairwise disjnt G and
  GF:  $G \subseteq F$ 
  by blast
define A where A = ( $\bigcup G$ )
from Suc(3) have card F > 0 by simp
hence finite F by (rule card-ge-0-finite)
from GF  $\langle \text{finite } F \rangle$  have finG: finite G by (rule finite-subset)
have card ( $\bigcup G$ )  $\leq$  sum card G
  using card-Union-le-sum-card by blast
also have ...  $\leq$  of-nat (card G) * Suc k
  by (metis GF Suc.prem1 le-Suc-eq subsetD sum-bounded-above)
also have ...  $\leq$  (r - 1) * Suc k
  using tr[folded cardG] by (metis id-apply mult-le-mono1 of-nat-eq-id)
finally have cardA: card A  $\leq$  (r - 1) * Suc k unfolding A-def .
{
  fix B
  assume *: B  $\in$  F
  with Suc(2) have nE: B  $\neq$  {} by auto
  from Suc(2) have eF: {}  $\notin$  F by auto
  have B  $\cap$  A  $\neq$  {}
  proof
    assume dis: B  $\cap$  A = {}
    hence disj: pairwise disjnt ({B}  $\cup$  G) using disj unfolding A-def
      by (smt (verit, ccfv-SIG) Int-commute Un-iff
        Union-disjoint disjnt-def pairwise-def singleton-iff)
    from nE dis have B  $\notin$  G unfolding A-def by auto
    with finG have c: card ({B}  $\cup$  G) = Suc t by (simp add: cardG)
    have P (Suc t) unfolding P-def pd-sub-def
      by (intro exI[of - {B}  $\cup$  G], insert eF disj c * GF, auto)
    with greatest show False by force
  qed
} note overlap = this
have F  $\neq$  {} using Suc(2-) by auto

```

```

with overlap have Ane:  $A \neq \{\}$  unfolding A-def by auto
have finite A unfolding A-def using finG Suc(2-) GF by auto
let  $?g = \lambda B\ x. x \in B \cap A$ 
define f where  $f = (\lambda B. \text{SOME } x. ?g\ B\ x)$ 
have  $f \in F \rightarrow A$ 
proof
  fix B
  assume  $B \in F$ 
  from overlap[OF this] have  $\exists x. ?g\ B\ x$  unfolding A-def by auto
  from someI-ex[OF this] show  $f\ B \in A$  unfolding f-def by auto
qed
from pigeonhole-card[OF this <finite F> <finite A> Ane]
obtain a where  $a: a \in A$ 
and le:  $\text{card } F \leq \text{card } (f - \{a\} \cap F) * \text{card } A$  by auto
{
  fix S
  assume  $S \in F\ f\ S \in \{a\}$ 
  with someI-ex[of ?g S] a overlap[OF this(1)]
  have  $a \in S$  unfolding f-def by auto
} note FaS = this
let  $?F = \{S - \{a\} \mid S. S \in F \wedge f\ S \in \{a\}\}$ 
from cardA have  $((r - 1) \wedge^k * \text{fact } k) * \text{card } A \leq ((r - 1) \wedge^k * \text{fact } k) * ((r - 1) * \text{Suc } k)$ 
by simp
also have  $\dots = (r - 1) \wedge (\text{Suc } k) * \text{fact } (\text{Suc } k)$ 
by (metis (no-types, lifting) fact-Suc mult.assoc mult.commute of-nat-id power-Suc2)
also have  $\dots < \text{card } (f - \{a\} \cap F) * \text{card } A$ 
using Suc(3) le by auto
also have  $f - \{a\} \cap F = \{S \in F. f\ S \in \{a\}\}$  by auto
also have  $\text{card } \dots = \text{card } ((\lambda S. S - \{a\}) ' \{S \in F. f\ S \in \{a\}\})$ 
by (subst card-image; intro inj-onI refl, insert FaS) auto
also have  $(\lambda S. S - \{a\}) ' \{S \in F. f\ S \in \{a\}\} = ?F$  by auto
finally have lt:  $(r - 1) \wedge^k * \text{fact } k < \text{card } ?F$  by simp
have  $\forall A \in ?F. \text{finite } A \wedge \text{card } A = k$  using Suc(2) FaS by auto
from Suc(1)[OF this lt] obtain S
where sunflower S  $\text{card } S = r\ S \subseteq ?F$  by auto
from  $\langle S \subseteq ?F \rangle\ \text{FaS}$  have  $S \subseteq \{A - \{a\} \mid A. A \in F \wedge a \in A\}$  by auto
from sunflower-remove-element-lift[OF this <sunflower S> <card S = r>]
show ?thesis by auto
qed
qed

```

Using *sunflower-card-subset-lift* we can easily replace the condition that the cardinality is exactly k by the requirement that the cardinality is at most k . However, then $\{\} \notin S$ cannot be ensured. Consider $r = 1 \wedge 0 < k \wedge F = \{\{\}\}$.

lemma *Erdos-Rado-sunflower*:

assumes $\forall A \in F. \text{finite } A \wedge \text{card } A \leq k$

and $\text{card } F > (r - 1)^k * \text{fact } k$
shows $\exists S. S \subseteq F \wedge \text{sunflower } S \wedge \text{card } S = r$
by (*rule sunflower-card-subset-lift*[*OF - assms*],
metis Erdos-Rado-sunflower-same-card)

We further provide a lower bound on the existence of sunflowers, i.e., Exercise 6.2 of the textbook [3]. To be more precise, we prove that there is a set of sets of cardinality $(r - 1)^k$, where each element is a set of cardinality k , such that there is no subset which is a sunflower with cardinality of at least r .

lemma *sunflower-lower-bound*:

assumes *inf*: *infinite* (*UNIV* :: 'a set)
and *r*: $r \neq 0$
and *rk*: $r = 1 \implies k \neq 0$
shows $\exists F.$
 $\text{card } F = (r - 1)^k \wedge \text{finite } F \wedge$
 $(\forall A \in F. \text{finite } (A :: 'a \text{ set}) \wedge \text{card } A = k) \wedge$
 $(\nexists S. S \subseteq F \wedge \text{sunflower } S \wedge \text{card } S \geq r)$
proof (*cases* $r = 1$)
case *False*
with *r* **have** *r*: $r > 1$ **by** *auto*
show *?thesis*
proof (*induct* *k*)
case 0
have *id*: $S \subseteq \{\{\}\} \longleftrightarrow (S = \{\} \vee S = \{\{\}\})$ **for** *S* :: 'a set **set** **by** *auto*
show *?case* **using** *r*
by (*intro exI*[*of* - $\{\{\}\}$], *auto simp*: *id*)
next
case (*Suc k*)
then obtain *F* **where**
 $\text{card } F = (r - 1)^k$ **and**
 $\text{finite } F$ **and**
 $\bigwedge A. (A :: 'a \text{ set}) \in F \implies \text{finite } A \wedge \text{card } A = k$ **and**
 $\neg (\exists S \subseteq F. \text{sunflower } S \wedge r \leq \text{card } S)$
by *metis*

main idea: get $k - 1$ fresh elements and add one of these to all elements of *F*

have *finite* $(\bigcup F)$ **using** *fin AF* **by** *simp*
hence *infinite* $(\text{UNIV} - \bigcup F)$ **using** *inf* **by** *simp*
from *infinite-arbitrarily-large*[*OF this*, *of* $r - 1$]
obtain *New* **where** *New*: *finite* *New* $\text{card } New = r - 1$
 $New \cap \bigcup F = \{\}$ **by** *auto*
define *G* **where** $G = (\lambda (A, a). \text{insert } a A) ` (F \times New)$
show *?case*
proof (*intro exI*[*of* - *G*] *conjI*)
show *finite* *G* **using** *New fin* **unfolding** *G-def* **by** *simp*
have $\text{card } G = \text{card } (F \times New)$ **unfolding** *G-def*
proof ((*subst card-image*; (*intro refl*)?), *intro inj-onI*, *clarsimp*, *goal-cases*)

```

case (1 A a B b)
hence ab:  $a = b$  using New by auto
from 1(1) have  $\text{insert } a \ A - \{a\} = \text{insert } b \ B - \{a\}$  by simp
also have  $\text{insert } a \ A - \{a\} = A$  using New 1 by auto
also have  $\text{insert } b \ B - \{a\} = B$  using New 1 ab[symmetric] by auto
finally show ?case using ab by auto
qed
also have  $\dots = \text{card } F * \text{card } \text{New}$  using New fin by auto
finally show  $\text{card } G = (r - 1) \wedge \text{Suc } k$ 
  unfolding cardF New by simp
{
  fix B
  assume  $B \in G$ 
  then obtain a A where  $G: a \in \text{New } A \in F \ B = \text{insert } a \ A$ 
    unfolding G-def by auto
  with AF[of A] New have finite B card B = Suc k
    by (auto simp: card-insert-if)
}
thus  $\forall A \in G. \text{finite } A \wedge \text{card } A = \text{Suc } k$  by auto
show  $\neg (\exists S \subseteq G. \text{sunflower } S \wedge r \leq \text{card } S)$ 
proof (intro notI, elim exE conjE)
  fix S
  assume *:  $S \subseteq G \text{ sunflower } S \ r \leq \text{card } S$ 
  define g where  $g \ B = (\text{SOME } a. a \in \text{New} \wedge a \in B)$  for B
  {
    fix B
    assume  $B \in S$ 
    with  $\langle S \subseteq G \rangle$  have  $B \in G$  by auto
    hence  $\exists a. a \in \text{New} \wedge a \in B$  unfolding G-def by auto
    from someI-ex[OF this, folded g-def]
    have  $g \ B \in \text{New } g \ B \in B$  by auto
  } note gB = this
  have  $\text{card } (g \ S) \leq \text{card } \text{New}$ 
    by (rule card-mono, insert New gB, auto)
  also have  $\dots < r$  unfolding New using r by simp
  also have  $\dots \leq \text{card } S$  by fact
  finally have  $\text{card } (g \ S) < \text{card } S$  .
  from pigeonhole[OF this] have  $\neg \text{inj-on } g \ S$  .
  then obtain B1 B2 where B12:  $B1 \in S \ B2 \in S \ B1 \neq B2 \ g \ B1 = g \ B2$ 
    unfolding inj-on-def by auto
  define a where  $a = g \ B2$ 
  from B12 gB[of B1] gB[of B2] have a:  $a \in \text{New } a \in B1 \ a \in B2$ 
    unfolding a-def by auto
  with B12 have  $\exists B1 \ B2. B1 \in S \wedge B2 \in S \wedge B1 \neq B2 \wedge a \in B1 \wedge a \in$ 
B2
    unfolding a-def by blast
  from  $\langle \text{sunflower } S \rangle$  [unfolded sunflower-def, rule-format, OF this]
  have aS:  $B \in S \implies a \in B$  for B by auto
  define h where  $h \ B = B - \{a\}$  for B

```

```

define T where T = h ` S
have  $\exists S \subseteq F. \text{sunflower } S \wedge r \leq \text{card } S$ 
proof (intro exI[of - T] conjI)
{
  fix B
  assume B  $\in S$ 
  have hB:  $h B = B - \{a\}$ 
  unfolding h-def T-def by auto
  from aS  $\langle B \in S \rangle$  have aB:  $a \in B$  by auto
  from  $\langle B \in S \rangle \langle S \subseteq G \rangle$  obtain a' A where AF:  $A \in F$ 
  and B:  $B = \text{insert } a' A$ 
  and a':  $a' \in \text{New}$  unfolding G-def by force
  from aB B a' New AF a(1) hB AF have insert a  $(h B) = B$  h B = A
by auto
  hence insert a  $(h B) = B$  h B  $\in F$  insert a  $(h B) \in S$  using AF  $\langle B \in S \rangle$  by auto
} note main = this
have CTS:  $C \in T \implies \text{insert } a C \in S$  for C using main unfolding
T-def by force
show  $T \subseteq F$  unfolding T-def using main by auto
have  $r \leq \text{card } S$  by fact
also have ... = card T unfolding T-def
by (subst card-image, intro inj-on-inverseI[of - insert a], insert main,
auto)
finally show  $r \leq \text{card } T$  .
show sunflower T unfolding sunflower-def
proof (intro allI impI, elim exE conjE, goal-cases)
case (1 x C C1 C2)
from CTS[OF  $\langle C1 \in T \rangle$ ] CTS[OF  $\langle C2 \in T \rangle$ ] CTS[OF  $\langle C \in T \rangle$ ]
have *: insert a C1  $\in S$  insert a C2  $\in S$  insert a C  $\in S$  by auto
from 1 have insert a C1  $\neq$  insert a C2 using main
unfolding T-def by auto
hence  $\exists A B. A \in S \wedge B \in S \wedge A \neq B \wedge x \in A \wedge x \in B$ 
using * 1 by auto
from  $\langle \text{sunflower } S \rangle [\text{unfolded sunflower-def, rule-format, OF this } *(3)]$ 
have  $x \in \text{insert } a C$  .
with 1 show  $x \in C$  unfolding T-def h-def by auto
qed
qed
with sf
show False ..
qed
qed
qed
next
case r: True
with rk have  $k \neq 0$  by auto
then obtain l where  $k = \text{Suc } l$  by (cases k, auto)
show ?thesis unfolding r k

```

by (intro exI[of - {}], auto)
qed

The difference between the lower and the upper bound on the existence of sunflowers as they have been formalized is *fact k*. There is more recent work with tighter bounds [1], but we only integrate the initial result of Erdős and Rado in this theory.

We further provide the Erdős Rado lemma lifted to obtain non-empty cores or cores of arbitrary cardinality.

lemma *Erdos-Rado-sunflower-card-core:*
assumes *finite E*
and $\forall A \in F. A \subseteq E \wedge s \leq \text{card } A \wedge \text{card } A \leq k$
and $\text{card } F > (\text{card } E \text{ choose } s) * (r - 1) \wedge k * \text{fact } k$
and $s \neq 0$
and $r \neq 0$
shows $\exists S. S \subseteq F \wedge \text{sunflower } S \wedge \text{card } S = r \wedge \text{card } (\bigcap S) \geq s$
by (rule *sunflower-card-core-lift*[OF *assms*(1) - *assms*(2) - *assms*(4-5),
of $(r - 1) \wedge k * \text{fact } k$],
rule *Erdos-Rado-sunflower*, insert *assms*(3), auto simp: *ac-simps*)

lemma *Erdos-Rado-sunflower-nonempty-core:*
assumes *finite E*
and $\forall A \in F. A \subseteq E \wedge \text{card } A \leq k$
and $\{\} \notin F$
and $\text{card } F > \text{card } E * (r - 1) \wedge k * \text{fact } k$
shows $\exists S. S \subseteq F \wedge \text{sunflower } S \wedge \text{card } S = r \wedge \bigcap S \neq \{\}$
by (rule *sunflower-nonempty-core-lift*[OF *assms*(1)
- *assms*(2-3), of $(r - 1) \wedge k * \text{fact } k$],
rule *Erdos-Rado-sunflower*, insert *assms*(4), auto simp: *ac-simps*)

end

References

- [1] Ryan Alweiss, Shachar Lovett, Kewen Wu, and Jiapeng Zhang. Improved bounds for the sunflower lemma. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing, STOC 2020*, pages 624–630. ACM, 2020. doi:10.1145/3357713.3384234.
- [2] Paul Erdős and Richard Rado. Intersection theorems for systems of sets. *Journal of the London Mathematical Society*, 35:85–90, 1960. doi:10.1112/jlms/s1-35.1.85.
- [3] Stasys Jukna. *Extremal Combinatorics*. Texts in Theoretical Computer Science. An EATCS Series. Springer, 2011. doi:10.1007/978-3-642-17364-6_6.